Phase portraits for distinct real Eigenvalues

When we learn to solve systems of differential equations, we discover that much of what we learned about solving individual differential equations, also extends to solving systems.

This is also true when it comes to sketching solution curves for the system. In the same way that we previously learned to identify equilibrium solutions for a differential equation, and then sketch solution curves through the direction field around those equilibrium solutions, we can also identify equilibrium solutions for a system of differential equations, and then sketch solution curves through the direction field around those equilibrium solutions.

Equilibrium solutions

For any homogeneous system of two differential equations $\vec{x}' = A\vec{x}$, the zero vector $\vec{x} = \vec{O}$,

$$\vec{x} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

is always a solution. The zero vector $\vec{x} = \vec{O}$ is an equilibrium solution because any vector satisfying $A\vec{x} = \vec{O}$ is an **equilibrium solution** to the system.

If the matrix A is nonsingular (A is a square matrix with a defined inverse), then $\vec{x} = \vec{O}$ will be the only equilibrium solution of the system. And in that case, the only question left to answer is how the solution curves,

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$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

are behaving around this single equilibrium solution. We'll need to consider both their shape and direction.

Phase plane, trajectories, and the phase portrait

When we dealt with solution curves for a single differential equation, we would sketch the curves $y(t)$ in the yt-plane. But now that we're dealing with vectors, we'll use the x_1x_2-plane.

In other words, we'll use the values from the solution vector, x_1 and x_2, to define a new plane, which we'll call the **phase plane**. We'll sketch **trajectories** in the phase plane, which are just specific solution curves. A picture of a collection of trajectories gives us the **phase portrait** of the system.

Fortunately, we already have everything we need to sketch phase portraits, because the phase portrait is dictated by the Eigenvalues and Eigenvectors of the system, and we already know how to find those.

Distinct real Eigenvalues

Here's what we want to notice about the phase portraits of systems with distinct real Eigenvalues.

1. The two straight lines correspond to the Eigenvectors of the system. An Eigenvector $\vec{k_1} = (k_1, k_2)$ will lie along the line $y = (k_2/k_1)x$.

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2. The direction along those linear trajectories is dependent on the sign of the associated Eigenvalue. A linear trajectory associated with a positive Eigenvalue will always move away from the origin, while a linear trajectory associated with a negative Eigenvalue will always move toward the origin.

3. The equilibrium of two positive Eigenvalues is an unstable node that repels all trajectories, while the equilibrium of two negative Eigenvalues is an asymptotically stable node that attracts all trajectories. The equilibrium of two opposite signed Eigenvalues is an unstable saddle point, because the trajectories move toward the origin at some points, but move away from it at others.

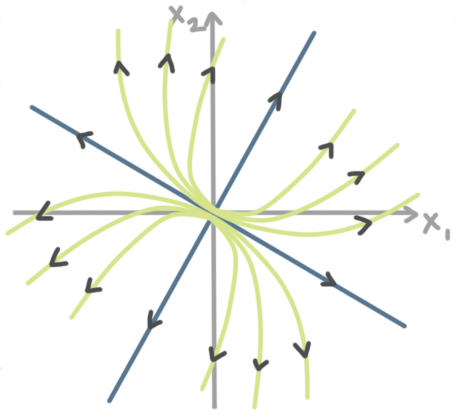
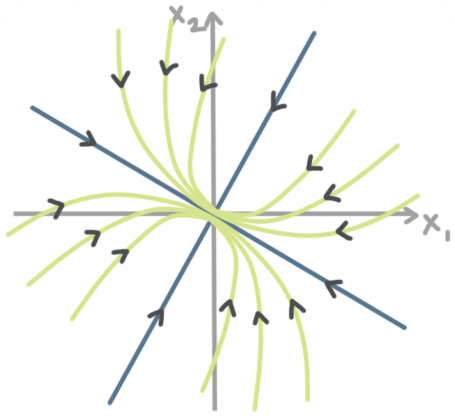
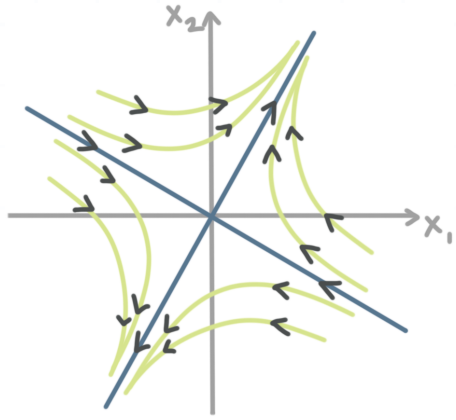
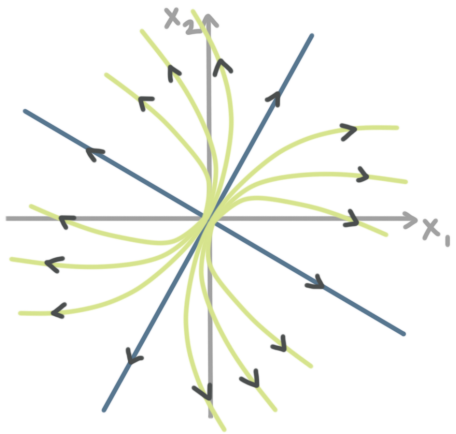
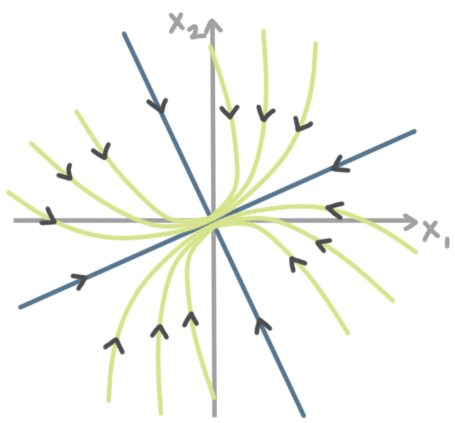
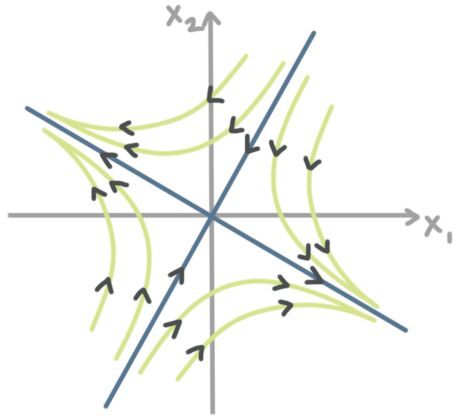
4. To check for the direction of the trajectories for any case of distinct real Eigenvalues, we can use the $t \to \pm\infty$ test. We can think about the trajectories "starting" at $t \to -\infty$, and "ending" at $t \to \infty$.

Below is a table that summarizes information about the phase portrait, based on the sign of the distinct real Eigenvalues.

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DISTINCT REAL EIGENVALUES

	Positive	Negative	Opposite signs
	$\lambda_1, \lambda_2 > 0$	$\lambda_1, \lambda_2 < 0$	$\lambda_1 > 0, \lambda_2 < 0$
Equilibrium	Node	Node	Saddle point
Stability	Unstable	Asymptotically stable	Unstable
	(Repeller)	(Attractor)	
Direction	$t \to \pm\infty$	$t \to \pm\infty$	$t \to \pm\infty$
Sketches			
			

Let’s do an example so that we can see how to build a phase portrait for distinct real Eigenvalues.

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Example

Sketch the phase portrait of the system.

$$x_1' = 2x_1 + 2x_2$$

$$x_2' = x_1 + 3x_2$$

The coefficient matrix and $A - \lambda I$ are,

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 2 \\ 1 & 3 - \lambda \end{bmatrix}$$

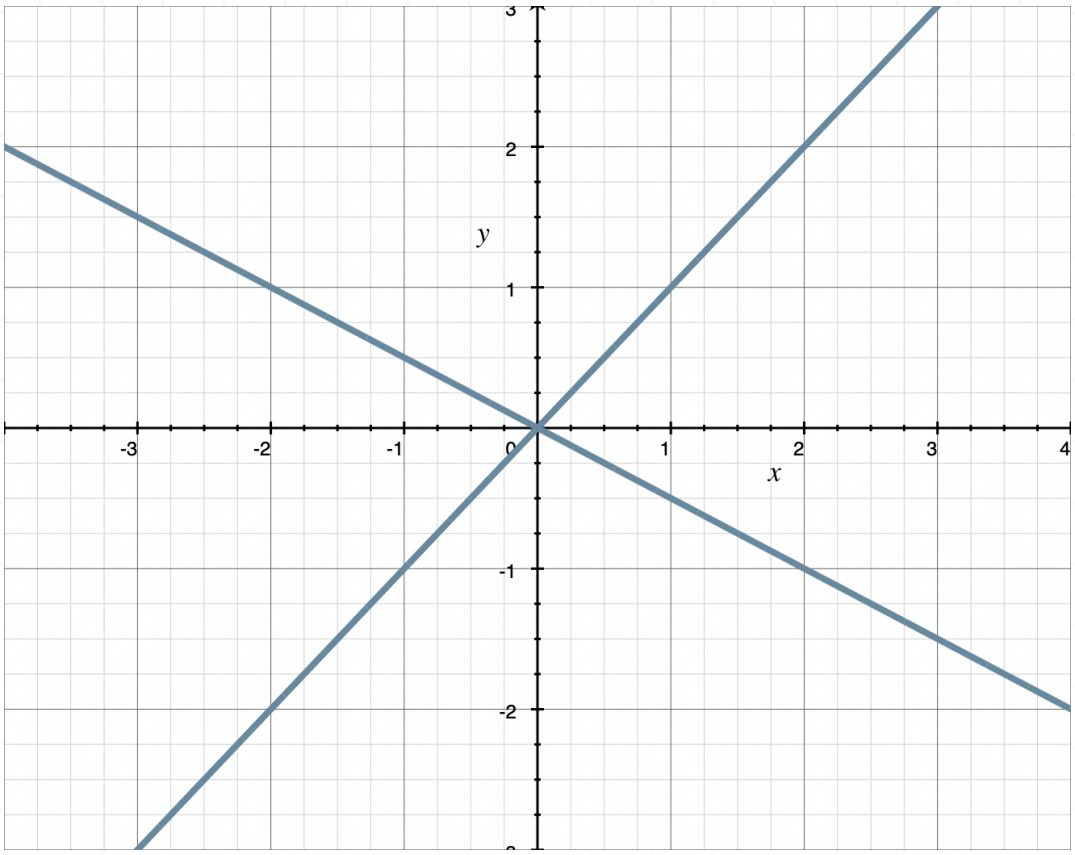
and the characteristic equation $\lambda^2 - 5\lambda + 4 = 0$ gives the Eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 4$, and their associated Eigenvectors

$$\overrightarrow{k_1} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The Eigenvector $\overrightarrow{k_1} = (-2,1)$ lies along the line $y = -(1/2)x$, and the Eigenvector $\overrightarrow{k_2} = (1,1)$ lies along the line $y = x$, so we'll sketch these lines.

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The Eigenvalue associated with $\overrightarrow{k_1} = (-2,1)$ is $\lambda = 1$, which means the direction along that trajectory is away the origin. The Eigenvalue associated with $\overrightarrow{k_2} = (1,1)$ is $\lambda = 4$, which means the direction along that trajectory is also away from the origin.

Because both Eigenvalues are positive, we're dealing with an unstable node that repels all trajectories.

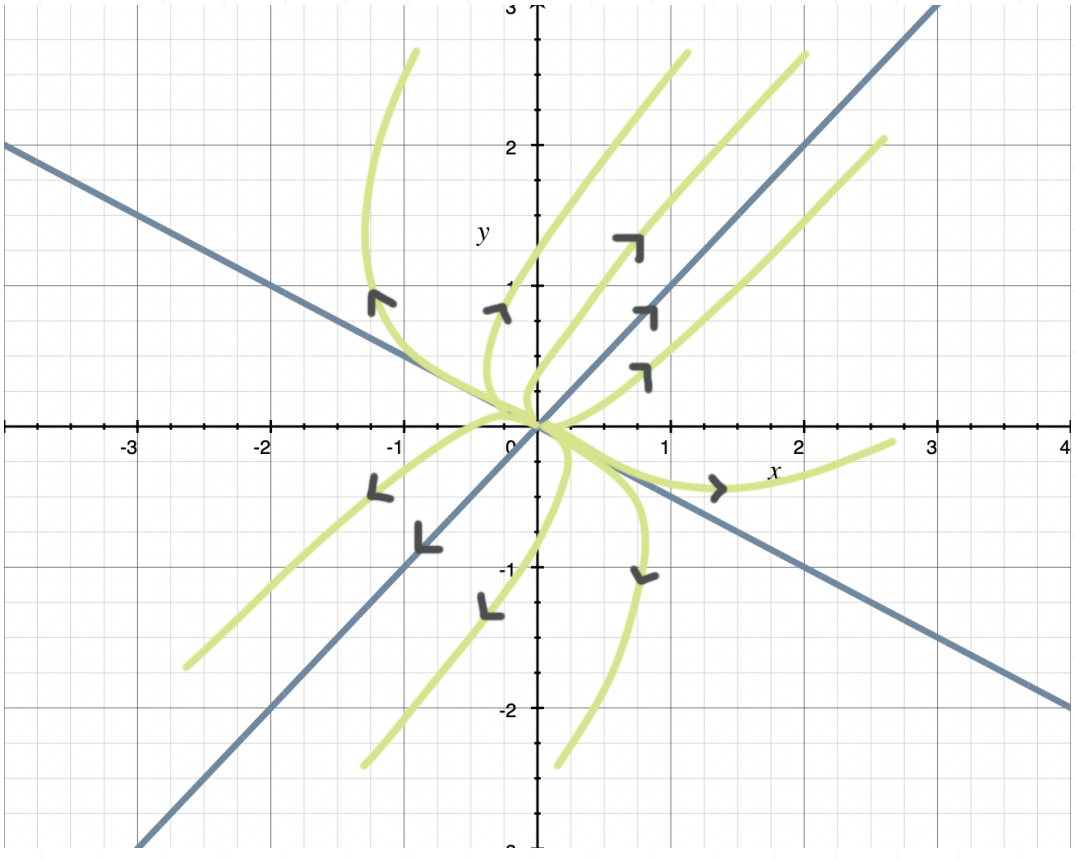
To apply the $t \to \pm\infty$ test, we'll use the general solution to the system,

$$\overrightarrow{x} = c_1 \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{4t}$$

Because e^{4t} goes to 0 faster than e^t as $t \to -\infty$, the $\overrightarrow{k_2} = (1,1)$ vector drops away first, meaning that our trajectories are going to "start" parallel to $\overrightarrow{k_1} = (-2,1)$. On the other end, e^{4t} dominates e^t as $t \to \infty$, so the $\overrightarrow{k_1} = (-2,1)$ vector will drop away first, meaning that our trajectories are going to "end" parallel to $\overrightarrow{k_2} = (1,1)$.

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Therefore, starting the trajectories parallel to $\overrightarrow{k_1} = (-2,1)$ and ending them parallel to $\overrightarrow{k_2} = (1,1)$ means that the phase portrait must look something like



And before we move on to looking at equal real Eigenvalues, let's do one more example with the special case where one Eigenvalue is $\lambda = 0$.

Example

Sketch the phase portrait of the system.

$$x_1' = -6x_1 + 2x_2$$

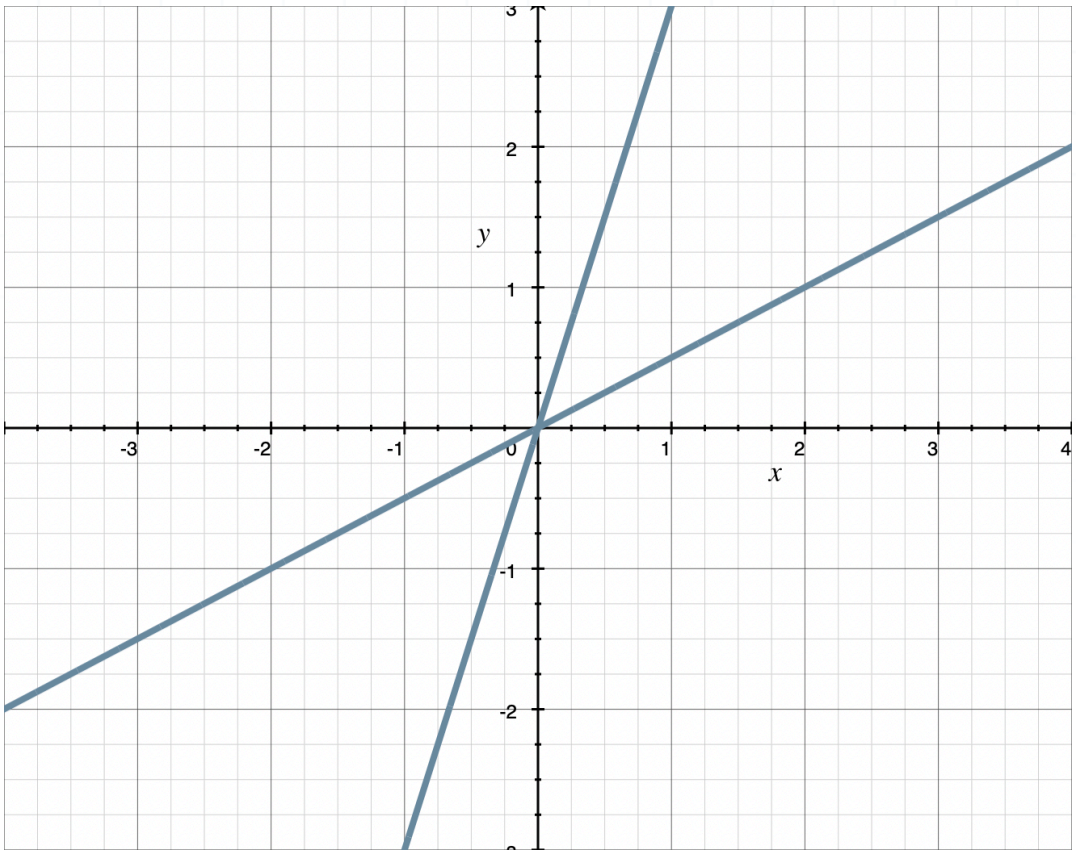
$$x_2' = -3x_1 + x_2$$

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This is the example we looked at when we learned to solve homogeneous systems with distinct real Eigenvalues, and we found that the Eigenvalues for the system were $\lambda = -5, 0$. The associated Eigenvectors were

$$\overrightarrow{k_1} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

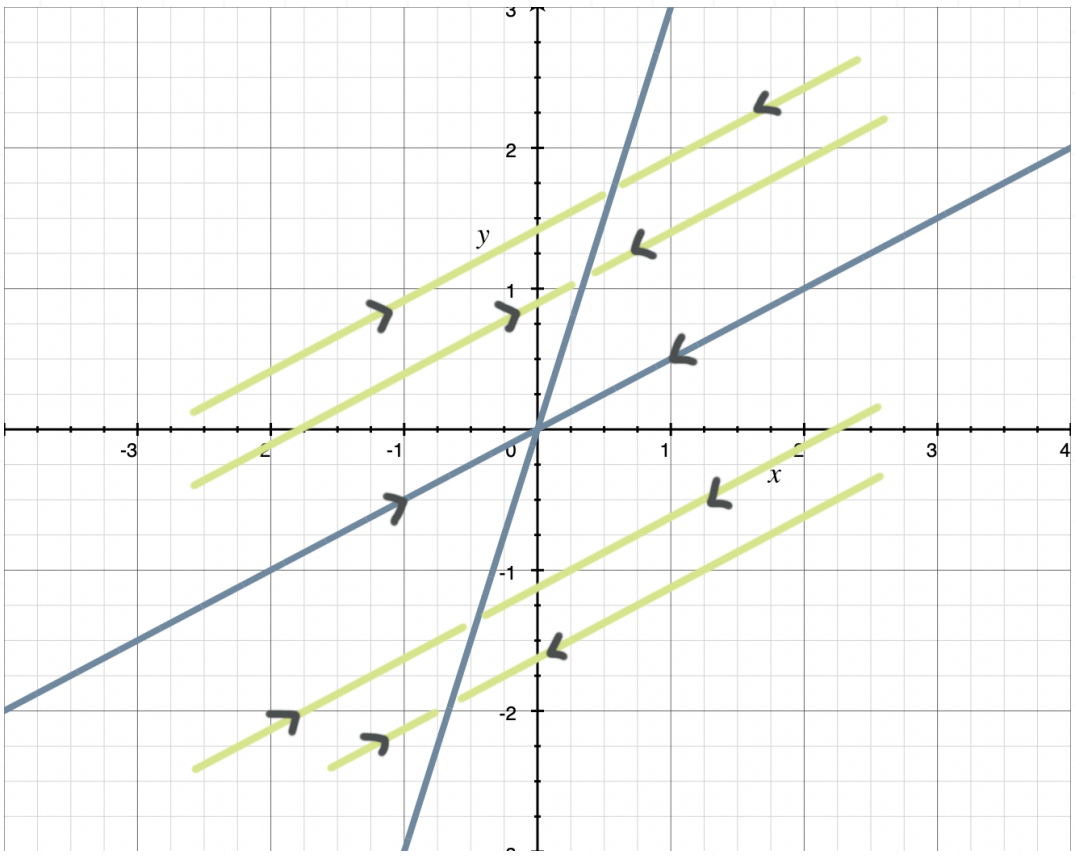
The Eigenvector $\overrightarrow{k_1} = (2,1)$ lies along the line $y = (1/2)x$, and the Eigenvector $\overrightarrow{k_2} = (1,3)$ lies along the line $y = 3x$, so we'll sketch these lines.



The Eigenvalue associated with $\overrightarrow{k_1} = (2,1)$ is $\lambda = -5$, which means the direction along that trajectory is toward the origin. The Eigenvalue associated with $\overrightarrow{k_2} = (1,3)$ is $\lambda = 0$, which means the direction along that trajectory is neither toward nor away from the origin. This is a special case in which equilibrium exists everywhere along $y = 3x$, not just at the origin (which occurs before of the special Eigenvalue $\lambda = 0$).

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Therefore, all trajectories run parallel to $y = (1/2)x$. Because $\lambda = -5$ is a negative Eigenvalue, the arrows always pointed toward $y = 3x$, and the stable line of equilibrium attracts all trajectories.



In other words, in the special case where one Eigenvalue is $\lambda = 0$,

- when the sign of the other Eigenvalue is negative, we'll have a stable line of equilibrium (attractor) along the Eigenvector associated with $\lambda = 0$, or
- when the sign of the other Eigenvalue is positive, we'll have an unstable line of equilibrium (repeller) along the Eigenvector associated with $\lambda = 0$.

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